

Efficient Dwelling—Small is Beautiful

by Shay Salomon

North Americans have quickly become the most-housed people in the history of humanity. The average detached single-family house built today has almost 2,500 square feet of floor area, and fewer than three people live in it. Each of these people has more than three times the space that new homeowners had in 1950.

In 1999 I saw a graph in *Environmental Building News* that showed what may be obvious to any HVAC professional. Energy use is primarily a function of the size (or more specifically, the volume) of the home. Technology can make houses more efficient per square foot, but what matters is absolute energy use. Reducing per capita living space is the single most effective way to reduce household energy use.

We Have Some Options

What can an individual builder do? If we work in new construction, especially with custom-built homes, we have an opportunity to influence homeowners to build smaller. There are so many reasons to do this that someone could write a book about it—everything from simple accounting (explain what each 100 square feet of floor area will cost the owner to heat, cool, and maintain over the life of the house) to aesthetic concerns (cut off some square feet and invest the money you save in more beautiful finishes) to technical design points (with a well-designed small house you can eliminate ductwork altogether, which generally increases the roof insulation and eliminates one maintenance chore).



NIGEL VALDEZ

This light-filled, open space is an example of gaining energy and space efficiency by combining a room's functions. The owner, pictured, lives very comfortably in a house one-sixth to one-tenth the size of her neighbors' houses in Oakland, California.

The mandate to legalize granny's

flat. In the past fifteen years, coalitions of transportation planners, environmentalists, architects, mass-transportation passengers, and home dwellers who would like to build structures in their yards or divide their house into two apartments, have come together in various parts of the country to legalize the construction and rental of "granny flats" (a.k.a. "coach houses," "mother-in-law apartments," "second units"; a.k.a. "accessory dwelling units," or "ADUs"). These small dwellings are either detached or attached to the main house, and usually inhabited by extended family or friends.

ADUs are a clear solution to a chain of problems: small families or single adults in large houses too expensive for even two-income couples to maintain happily by themselves; lack of reasonably priced rentals in some neighborhoods; separation of the generations; and the absence of sufficient urban density to operate cost-effective

public transportation. If they are placed near the back of a lot, ADUs can also add what some call "eyes on the alley," which is thought to discourage crime. ADUs are such a good idea that tens of thousands of otherwise law-abiding citizens built them illegally in the 1980s and '90s.

In the last two decades many cities have worked to legalize and promote ADUs, both to expand their source of tax revenue and to facilitate enforcement of fire and health codes. The State of California, in an attempt to create more affordable rentals, while lowering greenhouse gas emissions by discouraging sprawl, has enacted state legislation requiring municipalities to make it easier for owners to build ADUs. Dozens of cities and counties from Virginia and Massachusetts to Ontario and Idaho have passed similar ordinances. The American Association of Retired Persons (AARP) promotes ADUs, both as a way for grandchildren to live closer to grandparents, and as a source of post-retirement income: Grandma moves into her guesthouse, rents out her main house, and voilà, her house makes money instead of costing. Before you create an ADU, check your local zoning office. You may be a lucky resident of one of the many areas where they have always been legal.

More homes, less space. Cottage Housing ordinances are a solution in some areas. Some areas are platted for sprawling development, and zoned for low density. Homeowners in these areas currently assume that increased density in their neighborhood will harm them. But in areas where housing is in short supply, and citizens would like to preserve the last bits of forest and farmland, planners are looking for ways to encourage a little higher density. Some planning departments are encouraging "cluster developments" that construct houses closer together, then leave a larger, continuous, unbuilt area, sometimes as a shared park. A few municipalities have Cottage Housing ordinances

green building

which allow the construction of two, or sometimes just one-and-a-half small houses (meaning three on two lots) where otherwise one mansion might have been. In other places, on a limited number of lots per neighborhood, four or even six tiny homes are allowed where just one or two might have been before.

Codes and zoning laws are agreements among people about how to live. They were designed to protect tenants and buyers from callous landlords and developers, and to protect everyone from the dangers of the environment, built and natural. Almost all of us live within the code, but up to now a small group has determined the rules, defining for everyone what is a danger and what is safety. As more people with new perspectives enter the conversation about the code, maybe we'll turn our built environment into space that's more comfortable for all of us, present and future.

The option to subdivide. With existing construction, there is the parallel option to divide too-big

houses into apartments. Some real estate professionals have redirected their work toward helping groups of people buy large houses or small apartment complexes together, using tenants-in-common, cooperative, or community land trust models and trading the loneliness of a single-family home for the work of community building. Some builders have specialized their skills in this area, and can soundproof and fireproof adjacent walls, separate metering of energy use, and suggest designs that make higher-density to littler houses on a small planet living pleasant.

Other builders and designers work with more traditional single-family homeowners to develop their space—for example, for a growing family—without adding more volume to the house. Lofts, storage space under the floor and in unused corners, and well-placed new windows and window seats are a few of the many solutions they offer.

Social Fix to a Technological Problem

Homeowners may be less able to afford the newest technology at present—but they may have the time and motivation to make deep lifestyle changes that result in long-term energy gains. Builders, who have been accustomed to spending their time promoting products and technologies, are also in a good position to work as consultants, helping people adjust to these changes. **H_e**

*Shay Salomon is a carpenter and small-time homebuilder who coaches owner-builders who are trying to build mortgage-free. She is the author of **Little House on a Small Planet**, an informative, and inspiring book on living well in smaller houses.*

For more information:

Salomon, Shay. *Little House on a Small Planet*. The Lyons Press, 2006. To purchase the book, and for more helpful information about building smaller, go